

From Biological Question to Mathematical Problem

Research Process Essay (January 2025 - Present)

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I am drawn to systems that resist clean explanations. When a process is both highly studied and still difficult to explain, I instinctively want to understand why. This instinct has always pulled me toward problems where the answer is not obvious and there is no clear path to finding it.

The 18-month college-level Program for Algorithmic and Combinatorial Thinking introduced me to abstraction and formal reasoning. Almost three years on my school's competitive physics team trained me to build quantitative models from first principles, often from little more than a physical system, a set of observations, and a blank page. This trajectory culminated when I served as lead theorist for my team's first-place finish in the U.S. Invitational Young Physicists Tournament. Presenting and defending our reasoning before a panel of expert physicists and peers reinforced a lesson that became central to my research. Convincing explanations require clarity, sound technical work, and strong communication under critique.

Throughout high school, I was most interested in math, physics, and biology, but I found biological systems to be the most unforgiving. I became especially drawn to the immune system because it seemed difficult to reduce to any single mechanism, which led me to design an independent study in statistical immunology during my junior year. As I explored immune dysregulation with my teacher, rheumatoid arthritis presented a puzzle that refused to leave me alone. Why has RA incidence in the United States continued to rise since 1990 when two of its best-established risk factors, smoking and genetic susceptibility, have either declined or remained relatively stable at the population level? Meanwhile, obesity rates had increased sharply over the same period, and I began to wonder whether rising adiposity could be contributing to RA pathogenesis rather than merely coinciding with it.

When junior year ended, the independent study no longer provided a regular structure, but I continued pursuing the question on my own throughout the summer. I initially tried regression models to surface patterns, only to discover that young-onset RA datasets were limited, and more fundamentally, that observational data could not directly reconstruct exposure histories that might precede diagnosis by decades or reveal the counterfactual population outcomes I wanted to study. I eventually connected with Michael Bryan, an MD/PhD student at Oxford, who became my research mentor. Through our discussions, I realized early on that waiting for the 'perfect dataset' was itself a methodological trap. That realization forced me to rethink the problem and pivot toward simulation. I knew simulation could not recover exposure histories that had never been observed or prove causality from missing information, but it could make my assumptions explicit and let me test counterfactual scenarios under those assumptions.

Eighteen months in, my research has evolved through two stages and is now entering a third. Stage I moved from a biological question to a computational model built around stochastic simulation of long-latency disease risk. Stage II emerged when I began treating the model itself as something to understand. I started asking harder questions about what its assumptions allowed me to infer and where those assumptions might break down. Now, with recently secured access to longitudinal patient data, I am preparing to test the framework against real observations. Stage III is still taking shape, but I am most interested in whether its successes and failures reveal a deeper mathematical or statistical problem.

Stage I - From Biological Question to Computational Model

Once I committed to simulation, I began building the framework from the literature up. I built the simulation engine in Java as a Monte Carlo framework that generates synthetic populations of one million individuals, large enough to keep sampling noise small relative to the effects I wanted to examine. Rather than assigning fixed outcomes, I modeled individual risk probabilistically using published incidence estimates, odds ratios, and demographic factors, enabling me to compute prevalence patterns across age groups. I implemented an age-dependent exponential decay function to model attenuation of BMI-related risk over time and systematically perturbed obesity distributions to examine counterfactual population scenarios.

As an external consistency check, I compared predicted RA prevalence against Global Burden of Disease benchmarks and later used age-specific targets in calibration of selected model parameters. I also performed sensitivity analyses on key assumptions, including the BMI decay constant, to test whether plausible changes materially altered the conclusions. Each scenario was evaluated across 2,500 stochastic realizations, while uncertain parameters and assumptions were varied separately rather than treating repeated simulations as estimates of parameter uncertainty. This distinction became increasingly important as my work developed. As I refined the model, the differences between stochastic variability, parameter uncertainty, and model discrepancy became much clearer. Increasing the simulated population size or number of Monte Carlo realizations can reduce numerical uncertainty in estimated model outputs, but it cannot make uncertain parameters known or correct an inadequate model structure. That pushed me to think more carefully about the difference between uncertainty within the model and disagreement caused by the model itself.

That understanding did not come cleanly. Several early versions of the model produced implausible outputs and unstable results. As the implementation grew more complicated, the code also became increasingly difficult to debug. Because this work was not tied to a grade or

deadline, I was able to experiment freely, and that freedom proved essential. Rather than continuing to patch individual errors, I rebuilt the simulation several times from the ground up, making the framework increasingly modular and forcing myself to reconsider the assumptions embedded within it. In retrospect, many of the most important advances in my work came from the model's failures because they exposed gaps in my own reasoning.

As the model improved, my discussions with Michael helped me gradually constrain a plausible range for BMI's modeled population-level contribution to RA prevalence. Using Java for simulation and Python for analysis and visualization, I moved from rebuilding the framework to more targeted calibration and sensitivity analysis. Once the outputs stabilized within epidemiologically plausible bounds, the research began to feel less like constructing a model and more like understanding what the model could actually support.

That shift forced me to confront a broader methodological issue. I had built a simulator that could reproduce plausible population-level patterns, but I began to question what that agreement actually told me about the underlying system. Long-latency exposures are especially difficult because the relevant histories may be incomplete or entirely unobserved. Simulation cannot recover that missing information, and its conclusions remain conditional on the risk relationships, exposure distributions, and structural assumptions supplied to the model. What the model can do is make those assumptions explicit and test their consequences through counterfactual scenarios. But even a simulator that reproduces an observed prevalence curve is not necessarily the correct model of the underlying process, because different parameter combinations, exposure histories, or model structures may produce similar observable behavior. That distinction began to shift my attention from whether the model worked to what the model could actually allow me to infer.

After several months of revising the model, I began drafting a manuscript and submitted an abstract to the American Association of Immunologists Midwinter Conference. After scientific review, I was accepted to present and was the conference's only high school student presenter. Preparing the work for a professional audience forced me to compress months of modeling decisions into a clear argument, which sharpened both my conclusions and my understanding of the framework's assumptions and limitations. At the conference, I presented the model to clinical researchers and fielded questions about its structure, interpretation, and potential beyond rheumatoid arthritis. The experience was transformative as it pushed me to think more carefully about which aspects of the work were specific to the biological application and which might reflect a broader methodological problem.

Stage II — From Computational Model to Inference Problem

The conference also clarified that the framework was ready to be tested against real patient data. I posted the manuscript as a first-author preprint and began reaching out to clinical researchers to request access to longitudinal cohorts. After numerous proposals, rejections, calls, and revisions, I received approval to work with de-identified patient data from Seattle Children's Hospital and the restricted ERAS-ERAN cohorts through King's College London. I ultimately centered the next stage of my research on ERAS-ERAN, where my research proposal was reviewed and approved by the cohort's Steering Group.

Securing that access has been pivotal. What I had originally planned as a final validation step has become an opportunity to test not only whether the model matches real observations, but where its assumptions, parameterization, and inferential structure hold up and where they begin to break down. Just as importantly, I've come to realize that disagreement with the data would not by itself reveal *why* the model might fail. It could reflect poorly estimated parameters, stochastic variation, measurement error, omitted structure, or broader model misspecification. Distinguishing among those possibilities is now the central question driving the next stage of my research.

After graduation, the summer again gave me the freedom to work without the constraints of a busy school schedule. As I prepared to work with the longitudinal data, I began treating the framework less as a biological simulator and more as a mathematical object in its own right. I started asking what the model was actually doing as an inference system, which quantities were observed, which remained latent, and how assumptions about exposure histories and risk relationships entered the model.

Framing the model this way changed the questions I was asking. Instead of only asking whether the simulator could reproduce plausible population-level behavior, I began asking what information about the underlying states and parameters could actually be recovered from partial longitudinal observations, and under what assumptions that recovery would be identifiable and stable. If distinct latent histories, parameter combinations, or model structures produce essentially the same observations, additional computation alone cannot determine which generated the real system.

As that shift in my thinking became more concrete, I separated the new work from the original simulation codebase and created a second repository, *longitudinal-risk-inference*, focused on the new inference questions. I kept the original repository intact as a record of the completed simulator, while the new one became the workspace for testing the framework against longitudinal observations and developing the mathematical tools needed to understand where it succeeds and fails.

This new direction has pushed me toward questions of uncertainty, model discrepancy, and robustness. I am especially interested in separating uncertainty about unknown parameters from structural error in the model itself, since even a precisely estimated parameter would mean little

if the underlying model was wrong. I am also thinking more carefully about which parts of the framework should remain mechanistic and which, once the approved data are transferred for analysis, might instead be estimated or learned from data. The question driving my research has changed. I am no longer only asking whether the simulation can reproduce a plausible biological pattern. I want to know what can actually be inferred from partial observations of a stochastic system, under what assumptions that inference is identifiable and stable, and where the framework's limitations might reveal a more general mathematical or statistical problem.

While waiting to begin analysis of the patient cohorts, I have started building the computational and statistical framework needed for that transition. I redesigned the simulator as a longitudinal population process, with annual incidence, mortality, evolving BMI trajectories, and individual-level outcomes rather than relying primarily on static population summaries. This has given me a controlled environment in which the data-generating process is known and I can test inference procedures before applying them to real observations.

One preliminary experiment asks whether a flexible machine-learning model can identify structure that I deliberately encoded in the simulator without being told that structure explicitly. I trained a gradient-boosted model on simulated individual-level outcomes and used feature and interaction analyses to test whether it rediscovered known risk relationships, including the model's age-dependent BMI effect. Because the training data are generated by my own simulator, successful recovery is not biological evidence. Rather, it is a controlled test of whether the inference pipeline can recover known structure. The more interesting question is what happens when the same framework is applied to real longitudinal data, where the generating mechanism is unknown and the model itself may be wrong.

An assumption I had previously treated as necessary also became a research question in its own right. Because the original simulator lacked sufficient individual-level data, many relationships had to be specified from the literature. Longitudinal observations create the possibility of reversing that choice. Instead of assuming every relationship in advance, I can ask which structures the data themselves support. This has made me interested not simply in applying machine learning to the problem, but in understanding where learned components belong within a mechanistic model and what is lost or gained when a specified relationship is replaced by one inferred from data.

This reframing also required changing the simulator itself. The original framework was designed primarily to estimate population-level counterfactual differences. The new framework follows individuals longitudinally through annual transitions, allowing exposures and risk states to evolve before disease onset and producing individual trajectories that can eventually be compared with longitudinal observations. Rebuilding the model this way made the distinction between latent states and observable outcomes much more explicit and gave the inference questions a concrete mathematical form.

In the RA model, for example, a change in the assumed BMI-risk relationship and a change in the underlying BMI trajectory could potentially produce similar population-level incidence patterns. Observing the same outcome therefore may not be enough to determine which part of the generating process was responsible. Longitudinal individual-level measurements may constrain that ambiguity, but whether they constrain it enough is itself an inference question.

Stage III - From Forward Modeling to Learning from Observations

The next stage of my research is to use real longitudinal observations to find where the framework's assumptions or structure begin to break down. Comparing the simulator against ERAS-ERAN should help me determine whether discrepancies arise from uncertain parameters, stochastic variation, weakly identifiable latent structure, model misspecification, or relationships that would be better estimated from data than imposed mechanistically.

These possibilities are related, but they imply very different problems. If disagreement with the data can be explained by uncertainty in the parameters or stochastic variation, the existing model may still be adequate. If it instead reflects missing structure or systematic model discrepancy, then improving parameter estimates will not solve the problem. And if the available observations cannot distinguish between different latent histories or parameter values in the first place, then the limitation is one of identifiability rather than computation. Learning to tell these possibilities apart is now one of the central challenges of my research.

I do not yet know which of these limitations will prove most important, and I do not want to choose a mathematical method before the data reveal what problem actually needs to be solved. My first task is therefore to determine what the longitudinal observations can actually tell me about the model. That might expose an identifiability problem, reveal systematic model discrepancy, or show that certain mechanistic relationships would be better learned from data. It may also raise questions I have not anticipated. I want the mathematical direction of this stage to emerge from the limitations exposed by the data rather than forcing the data into a problem I have already decided to solve.

And even if the model fits the longitudinal data well, that alone will not establish that it has recovered the true underlying mechanism. Different latent processes may still produce similar observations, and some assumptions may remain impossible to test with the available data. I

therefore want to understand not only whether the model fits, but what aspects of the underlying system the data actually constrain and which conclusions remain dependent on assumptions that the observations cannot resolve.

What matters to me is not making the framework mathematically more elaborate for its own sake, but identifying a limitation whose resolution requires a genuinely new method or result. Rheumatoid arthritis gives me a concrete system in which those limitations can be exposed and tested, but I increasingly see the disease as a testbed rather than the endpoint of the research. If the same inferential difficulty arises from partial observation, latent structure, model misspecification, or the interaction between mechanistic and learned components, then the question may extend well beyond this particular disease. At the same time, I want this work to bring me closer to the question that originally motivated my research, whether rising adiposity could plausibly contribute to young-onset RA. My goal is to determine whether the model's successes and failures reveal such a problem and, if they do, formulate it precisely enough to study in its own right.

Looking Back to See Forward

After 18 months of sustained work, this process has fundamentally changed my understanding of both research and my own reasoning. I began with a question that felt logically inconsistent, assuming the primary challenge would be analytical. Instead, I encountered unstable models, flawed assumptions, data access barriers, and long periods of wondering where to go next. The process has made clear to me that meaningful research is not a linear progression of correct ideas, but an iterative system where errors expose hidden weaknesses and force clearer thinking.

What remains constant is my own drive to keep working on a problem that refuses to resolve cleanly. The experience of staying with a question long enough to reconstruct the way it must be approached has become central to how I engage with complex systems. Most importantly, my drive to solve complex problems comes from the simple, stubborn desire to understand systems that resist easy explanation.

The most important transition in my research has been from using a model to produce an answer to using the model to discover what I do not yet know how to answer. Building larger simulations initially seemed like a route toward greater certainty. Instead, the model's limitations exposed questions about identifiability, model discrepancy, partial observation, and the boundary between assumed and learned structure. As I move forward, I want to develop the mathematics and statistics needed to turn those limitations into precise questions and, where possible, methods for answering them. My hope is that this work will ultimately bring me closer to answering my original question about whether adiposity could plausibly contribute to young-onset RA, while also developing ideas that may apply to other long-latency diseases.